

Tutorial – Isaac Lab

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Isaac Lab

► #1 Basic Tutorial

- step-by-step guide to help you understand and use various features of the framework.

Setting up a Simple Simulation

These tutorials show you how to launch the simulation with different settings and spawn objects in the simulated scene. They cover the following APIs: `Applauncher`, `SimulationContext`, and `spawners`.

Creating an empty scene

Spawning prims into the scene

Deep-dive into Applauncher

Interacting with Assets

Having spawned objects in the scene, these tutorials show you how to create physics handles for these objects and interact with them. These revolve around the `AssetBase` class and its derivatives such as `RigidBody`, `Articulation` and `DeformableObject`.

Adding a New Robot to Isaac Lab

Interacting with a rigid object

Interacting with an articulation

Interacting with a deformable object

Interacting with a surface gripper

Creating a Scene

With the basic concepts of the framework covered, the tutorials move to a more intuitive scene interface that uses the `InteractiveScene` class. This class provides a higher level abstraction for creating scenes easily.

Using the Interactive Scene

Designing an Environment

The following tutorials introduce the concept of manager-based environments:

`ManagerBasedEnv` and its derivative `ManagerBasedREnv`, as well as the direct workflow base class `DirectREnv`. These environments bring-in together different aspects of the framework to create a simulation environment for agent interaction.

Creating a Manager-Based Base Environment

Creating a Manager-Based RL Environment

Creating a Direct Workflow RL Environment

Registering an Environment

Training with an RL Agent

Configuring an RL Agent

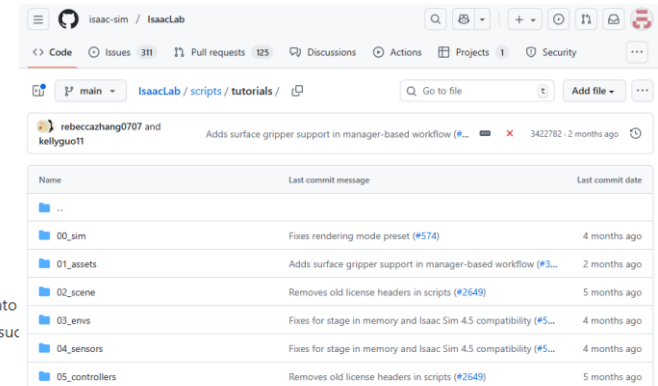
Modifying an existing Direct RL Environment

Policy Inference in USD Environment

Integrating Sensors

The following tutorial shows you how to integrate sensors into tutorials introduce the `SensorBase` class and its derivatives suc

Adding sensors on a robot



The screenshot shows the GitHub repository for IsaacLab. The file browser displays the following structure:

Name	Last commit message	Last commit date
..		
00_sim	Fixes rendering mode preset (#574)	4 months ago
01_assets	Adds surface gripper support in manager-based workflow (#3...	2 months ago
02_scene	Removes old license headers in scripts (#2649)	5 months ago
03_envs	Fixes for stage in memory and Isaac Sim 4.5 compatibility (#5...	4 months ago
04_sensors	Fixes for stage in memory and Isaac Sim 4.5 compatibility (#5...	4 months ago
05_controllers	Removes old license headers in scripts (#2649)	5 months ago

A recent commit by rebeccazhang0707 and kellyguo11 is highlighted: "Adds surface gripper support in manager-based workflow (#3422782) · 2 months ago".

Using motion generators

While the robots in the simulation environment can be controlled at the joint-level, the following tutorials show you how to use motion generators to control the robots at the task-level.

Using a task-space controller

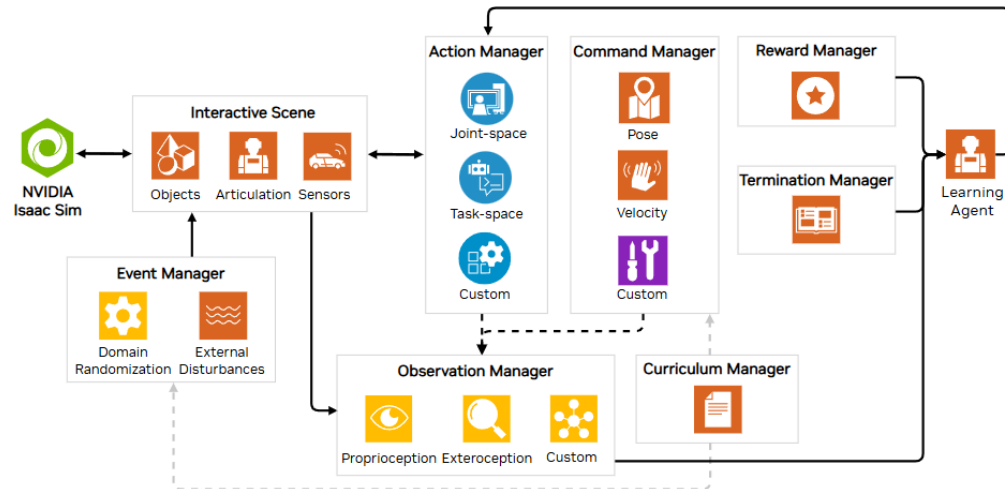
Using an operational space controller

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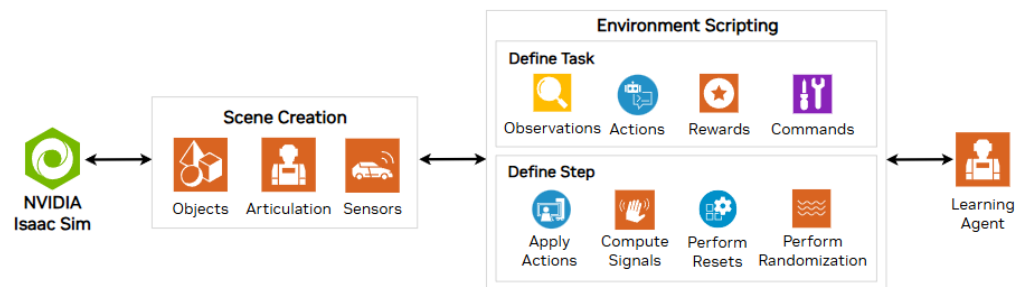
► #2 Robot Learning Framework

- Explain the two task design methods: Manager-Based & Direct
- analyze 3 examples for each.

Manager-based



Direct

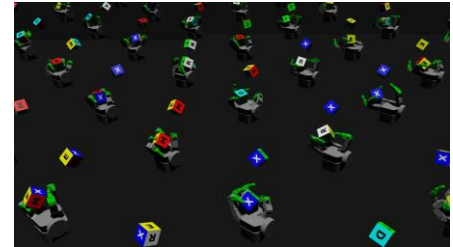
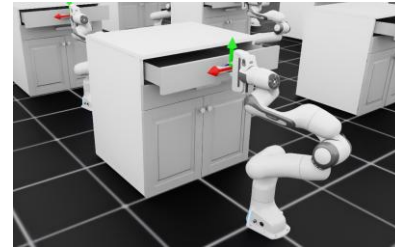
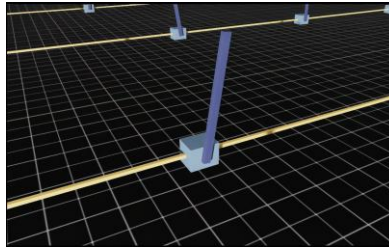


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► #2 Robot Learning Framework

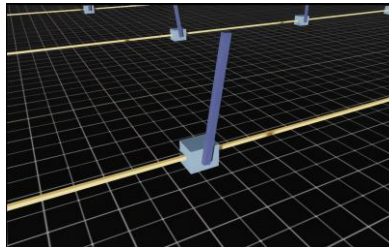
– Manager Based

- Cartpole
- Open-Drawer-Franka
- Repose-Cube-Allegro



– Direct

- Cartpole
- Humanoid



Isaac Lab

► #3 External Project

- Manager Based
 - Go2w
 - Custom Robot

► #4 Deploying policies in Isaac Sim

- Manager Based
 - Go2w
 - Custom Robot

► #5 Sim2Sim: Isaac Lab → Gazebo

- Manager Based
 - Go2w
 - Custom Robot



Isaac Lab

► How to Start?

- Github repository: https://github.com/chohh7391/isaacclab_tutorial
- Requirements
 - OS: Ubuntu 22.04
 - Mini Conda
 - Nvidia Graphic Driver
 - Cuda Toolkit (version > 12.8)

Thank You